Exercise Sheet 2 (Lectures 3 and 4)

Short-Answer Questions

Exercise 1 (Subspace):

What is the definition of a subspace $V \subseteq \mathbb{R}^m$?

Exercise 2 (Inverse Matrix):

Write down the inverse of

$$A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}.$$

Questions on Lecture 3

Exercise 3 (Extending a Basis):

Let $b_1, \ldots, b_n \in \mathbb{R}^m$ be a basis of V, and let $w \in \mathbb{R}^m \setminus V$. Show that $b_1, \ldots, b_n, w$ are linearly independent.

Exercise 4 (Full-Dimensional Subspaces):

Use the previous exercise to show that if V is an m-dimensional subspace of $\mathbb{R}^m$, then $V = \mathbb{R}^m$.

Exercise 5 (Existence of a Basis):

In the lecture, we showed that all bases have the same number of elements. However, we have not yet shown that a basis always exists. Use the previous exercises to prove that every subspace $V \subseteq \mathbb{R}^m$ has a basis.

Hint: Start with the empty set and repeatedly choose a vector of V outside the span of the vectors already chosen. Why must this process terminate?

Exercise 6 (Calculating a Null Space):

Find a basis of the null space

$$\mathcal{N}(A) = \{x \in \mathbb{R}^4 : Ax = 0\} \quad \text{for} \quad A = \begin{pmatrix} 1 & 2 & -1 & 0 \\ 2 & 5 & -1 & 2 \end{pmatrix}.$$

What is its dimension?

Exercise 7 (Intersection and Union):

Let U and V be subspaces of $\mathbb{R}^m$. Show that $U \cap V$ is a subspace. Is $U \cup V$ necessarily a subspace?

Questions on Lecture 4

Exercise 8 (Associativity and Distributivity):

Let

$$A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix}, \qquad C = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}.$$

Verify by direct calculation that

$$(AB)C = A(BC), \qquad A(B + C) = AB + AC.$$

Exercise 9 (Column Space of a Product):

Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times k}$. Show that

$$\mathcal{C}(AB) \subseteq \mathcal{C}(A).$$

Give an example where the inclusion is strict. Find a sufficient condition on B for equality.

Exercise 10 (Product of Invertible Matrices):

Let $A, B \in \mathbb{R}^{n \times n}$ be invertible. Show that AB is invertible and determine its inverse.

Exercise 11 (Lower-Triangular Matrices):

A matrix $A \in \mathbb{R}^{n \times n}$ is called *lower-triangular* if $a_{ij} = 0$ whenever $i < j$. Show that the product of two lower-triangular matrices is lower-triangular.

Exercise 12 (Column–Row Expansion):

Let $A_1, \ldots, A_k$ denote the columns of $A \in \mathbb{R}^{m \times k}$, and let $B^1, \ldots, B^k$ denote the rows of $B \in \mathbb{R}^{k \times n}$. Show that

$$AB = A_1 B^1 + \cdots + A_k B^k.$$

Here each product $A_\ell B^\ell$ is an $m \times n$ matrix.